Geologist James Hutton was the first to consider the age of the Earth scientifically. His fieldwork in the Scottish landscape revealed evidence to demonstrate that the Earth was far older than had been thought. His work laid the foundations for the modern discipline of geology.

James Hutton was born in Edinburgh to the family of a British merchant in 1726. After showing interest in mathematics and chemistry, he became a physician's assistant at 18 and went on to study medicine in Edinburgh, Paris, and the Netherlands. Rather than becoming a doctor, Hutton then set up a chemical plant and modernized his family's farms.

Hutton's fieldwork, allied with his agricultural and chemical knowledge, led him to lay the founding principles of geology. "Unconformity" was his term for the junction between different rock strata (layers) that had been laid down at different times. Unconformities can be seen in cliffs and outcrops as the rock layers above them differ from those below. Hutton also realized that rocks were formed under intense subterranean heat and pressure in a "uniformitarian" process, meaning that it occurs at the same rate today as it has always done. This made it possible to estimate the age of rock formations, and therefore the Earth, leading him to conclude that the Earth was far older than the 6,000 years proposed by biblical scholars.

MILESTONES

IN THE FIELD
Tours the north of Scotland in 1764 with George Clerk-Maxwell in search of geological features.

DIGGING DEEP
In 1767, studies rock strata revealed in excavations for the Forth and Clyde canal, Scotland.

UNIFORM THINKING
Publishes theory of uniformitarianism in a paper for the Royal Society of Edinburgh in 1785.

THEORY OF THE EARTH
Claims in 1788 that much of the land was once under the sea, and layers of rock are distorted over time.

***Hutton used** the term "unconformity" for a gap in the rock sequence at the point where the vertical layers meet the horizontal ones, as seen in his 1787 sketch of a rock face from Jedburgh, Scotland.*